

SOLUCIÓN TAREA 11

Paso 1: Instalación de Docker en el Host (Kali)

Instalamos y activamos Docker en la máquina virtual para descargar Ubuntu después.

- Actualizamos el sistema: apt update

```
root@kali:~# apt update
Get:1 https://mirrors.ubuntu.com/mirrors.txt [6.538 B]
Get:2 https://repos.azul.com/zulu/deb stable InRelease [6,538 B]
Get:3 http://kali.download/kali kali-rolling InRelease [26.8 kB]
Get:4 https://repos.azul.com/zulu/deb stable/main amd64 Packages [397 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Get:6 https://repos.azul.com/zulu/deb stable amd64 Contents (deb) [5.48 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Packages [128 kB]
Get:9 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [275 kB]
Get:10 http://kali.download/kali kali-rolling/non-free amd64 Packages [184 kB]
Get:11 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893 kB]
Get:12 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Get:13 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [33.8 kB]
Fetched 76.3 MB in 13s (6,909 kB/s)
2138 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Translations (main/i18n/translation-en-US) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Translations (main/i18n/translation-en) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Contents-deb (main/Contents-deb) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Translations (main/i18n/translation-en-US) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Translations (main/i18n/translation-en) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Contents-deb (main/Contents-deb) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
Warning: Target Contents-deb (main/Contents-all) is configured multiple times in /etc/apt/sources.list.d/zulu-openjdk.list:1 and /etc/apt/sources.list.d/zulu.list:1
```

- Instalamos Docker: apt install -y docker.io

```
root@kali:~# apt install -y docker.io
Upgrading:
libnl-3-200 libnl-genl-3-200 libnl-route-3-200 libperl5.40 perl perl-base perl-modules-5.40

Installing:
docker.io

Installing dependencies:
containerd criu docker-buildx docker-cli libcompel libintl-perl libintl-xs-perl libmodule-find-perl libnet9 libproc-processstable-perl libsort-naturally-perl needrestart python3-pycurl runc tini-static

Suggested packages:
containerd-plugin-plugins docker-doc btrfs-progs debootstrap rinse rootlesskit xfsprogs zfs-fuse | zfsutils-linux

Summary:
Upgrading: 7, Installing: 16, Removing: 0, Not Upgrading: 2131
Download size: 86.4 MB
Space needed: 326 MB / 58.6 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 runc amd64 1.3.5d01-1 [6,726 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libperl5.40 amd64 5.40.1-7 [4,317 kB]
Get:3 http://mirror.azul.com/mirrors/kali kali-rolling/main amd64 libcompel1 amd64 4.2-2 [63.0 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 perl amd64 5.40.1-7 [267 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 perl-base amd64 5.40.1-7 [1,679 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 perl-modules-5.40 all 5.40.1-7 [13,812 kB]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 libnet9 amd64 1.3.0dfsg-3 [51.3 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 libnl-route-3-200 amd64 3.12.0-2 [280 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 libnl-genl-3-200 amd64 3.12.0-2 [18.8 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 criu amd64 4.2-2 [556 kB]
Get:11 http://http.kali.org/kali kali-rolling/main amd64 docker-buildx amd64 9.29.1-d01-2 [14.1 MB]
Get:12 http://http.kali.org/kali kali-rolling/main amd64 docker-cli amd64 28.5.2-dfsg3-2 [7,884 kB]
Get:13 http://http.kali.org/kali kali-rolling/main amd64 containerd amd64 2.1.4-d02-0 [27.8 MB]
Get:14 http://http.kali.org/kali kali-rolling/main amd64 tini-static amd64 0.19.0-6-b1 [281 kB]
Get:15 http://http.kali.org/kali kali-rolling/main amd64 docker.io amd64 28.5.2-dfsg3-2 [18.5 MB]
Get:16 http://http.kali.org/kali kali-rolling/main amd64 libintl-perl all 3.37-1 [106 kB]
Get:17 http://kali.download/kali kali-rolling/main amd64 libmodule-find-perl all 0.17-1 [18.7 kB]
Get:18 http://http.kali.org/kali kali-rolling/main amd64 libproc-processstable-perl amd64 0.037-1+b1 [42.3 kB]
Get:19 http://mirror.es.cdn-perprod.com/kali kali-rolling/main amd64 libnl-3-200 amd64 3.12.0-2 [62.2 kB]
Get:20 http://mirror.es.cdn-perprod.com/kali kali-rolling/main amd64 libintl-xs-perl amd64 1.37-1 [16.9 kB]
Get:21 http://mirror.es.cdn-perprod.com/kali kali-rolling/main amd64 libsort-naturally-perl all 1.03-4 [13.1 kB]
Get:22 http://mirror.es.cdn-perprod.com/kali kali-rolling/main amd64 needrestart all 3.11-1 [68.6 kB]
Get:23 http://mirror.es.cdn-perprod.com/kali kali-rolling/main amd64 python3-pycurl all 4.2-2 [43.1 kB]
Fetched 86.4 MB in 21s (4,159 kB/s)
Reading database ... 419236 files and directories currently installed.)
Preparing to unpack .../libperl5.40_5.40.1-7_amd64.deb ...
Unpacking libperl5.40:amd64 (5.40.1-7) over (5.40.1-6) ...
Preparing to unpack .../perl_5.40.1-7_amd64.deb ...
Unpacking perl (5.40.1-7) over (5.40.1-6) ...
Preparing to unpack .../perl-base_5.40.1-7_amd64.deb ...
Unpacking perl-base (5.40.1-7) over (5.40.1-6) ...
Setting up perl-base (5.40.1-7) ...
```

- Iniciamos el servicio: systemctl start docker

```
root@kali:~# systemctl start docker
```

- Verificamos la instalación: Docker --version

```
root@kali:~# docker --version
Docker version 28.5.2+dfsg3, build 9cc6dea35e9a963f281434761c656fba4ac43aed
```

Paso 2: Creación de la “Máquina del Tiempo”

Descargamos Ubuntu 20.04 y entramos en ella: `Docker run -it --network host --name entorno-hue Ubuntu:20.04 /bin/bash`

```
(root@nodo1)-[~]  
# docker run -it --network host --name entorno-hue ubuntu:20.04 /bin/bash  
root@nodo1:/#
```

Al ejecutar el anterior comando el prompt de mi terminal cambia, el siguiente comando que escribimos es para instalar todas las dependencias del sistema y bibliotecas de desarrollo necesarias para compilar HUE.

Los comandos utilizados son:

- ```
- apt update
- export DEBIAN_FRONTEND=noninteractive
- apt install -y wget make ant gcc g++ rsync libffi-dev libgmp-dev \
python3-dev default-libmysqlclient-dev libkrb5-dev mysql-client \
ssl-dev libsasl2-dev libsasl2-modules-gssapi-mit libsqlite3-dev \
ldap2-dev libtidy-dev libxml2-dev libxslt1-dev python3-simplejson \
python3-setuptools python3-venv python-is-python3 virtualenv maven \
python3-pip nodejs npm nano
```

[illegible]

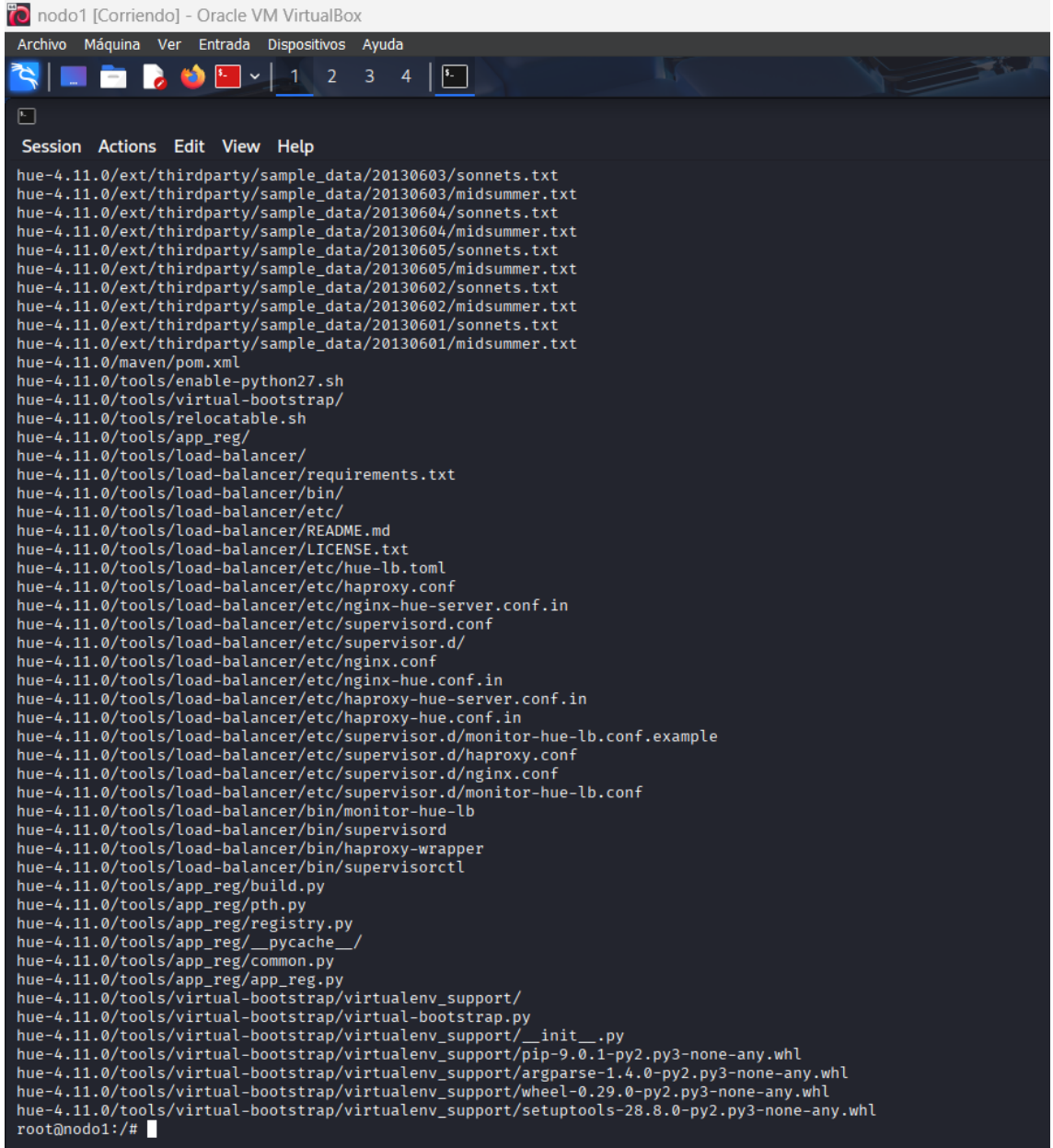
```
root@nodo1:/# export DEBIAN_FRONTEND=noninteractive
root@nodo1:/# apt install -y wget make ant gcc g++ rsync libffi-dev libgmp-dev \
> python3-dev default-libmysqlclient-dev libkrb5-dev mysql-client \
> libssl-dev libsasl2-dev libsasl2-modules-gssapi-mit libsqlite3-dev \
> libldap2-dev libtidy-dev libxml2-dev libxslt1-dev python3-simplejson \
> python3-setuptools python3-venv python-is-python3 virtualenv maven \
> python3-pip nodejs npm nano
```

### Paso 3: Descarga y Compilación de HUE

- Descargamos el código fuente: wget <https://cdn.gethue.com/downloads/hue-4.11.0.tgz>

```
root@nodo1:/# wget https://cdn.gethue.com/downloads/hue-4.11.0.tgz
--2026-04-12 15:18:13-- https://cdn.gethue.com/downloads/hue-4.11.0.tgz
Resolving cdn.gethue.com (cdn.gethue.com) ... 2606:4700:3032::6815:316a, 2606:4700:3035::ac43:bdbf, 172.67.189.191, ...
Connecting to cdn.gethue.com (cdn.gethue.com)|2606:4700:3032::6815:316a|:443 ...
```

- Descomprimos el archivo: tar xvf hue-4.11.0.tgz



The screenshot shows a terminal window titled 'nodo1 [Corriendo] - Oracle VM VirtualBox'. The terminal displays the output of the 'tar xvf hue-4.11.0.tgz' command, listing the contents of the extracted archive. The files are organized into several directories, including 'sample\_data', 'tools', and 'virtual-bootstrap'. The terminal output is as follows:

```
Session Actions Edit View Help
hue-4.11.0/ext/thirdparty/sample_data/20130603/sonnets.txt
hue-4.11.0/ext/thirdparty/sample_data/20130603/midsummer.txt
hue-4.11.0/ext/thirdparty/sample_data/20130604/sonnets.txt
hue-4.11.0/ext/thirdparty/sample_data/20130604/midsummer.txt
hue-4.11.0/ext/thirdparty/sample_data/20130605/sonnets.txt
hue-4.11.0/ext/thirdparty/sample_data/20130605/midsummer.txt
hue-4.11.0/ext/thirdparty/sample_data/20130602/sonnets.txt
hue-4.11.0/ext/thirdparty/sample_data/20130602/midsummer.txt
hue-4.11.0/ext/thirdparty/sample_data/20130601/sonnets.txt
hue-4.11.0/ext/thirdparty/sample_data/20130601/midsummer.txt
hue-4.11.0/maven/pom.xml
hue-4.11.0/tools/enable-python27.sh
hue-4.11.0/tools/virtual-bootstrap/
hue-4.11.0/tools/relocatable.sh
hue-4.11.0/tools/app_reg/
hue-4.11.0/tools/load-balancer/
hue-4.11.0/tools/load-balancer/requirements.txt
hue-4.11.0/tools/load-balancer/bin/
hue-4.11.0/tools/load-balancer/etc/
hue-4.11.0/tools/load-balancer/README.md
hue-4.11.0/tools/load-balancer/LICENSE.txt
hue-4.11.0/tools/load-balancer/etc/hue-lb.toml
hue-4.11.0/tools/load-balancer/etc/haproxy.conf
hue-4.11.0/tools/load-balancer/etc/nginx-hue-server.conf.in
hue-4.11.0/tools/load-balancer/etc/supervisord.conf
hue-4.11.0/tools/load-balancer/etc/supervisor.d/
hue-4.11.0/tools/load-balancer/etc/nginx.conf
hue-4.11.0/tools/load-balancer/etc/nginx-hue.conf.in
hue-4.11.0/tools/load-balancer/etc/haproxy-hue-server.conf.in
hue-4.11.0/tools/load-balancer/etc/haproxy-hue.conf.in
hue-4.11.0/tools/load-balancer/etc/supervisor.d/monitor-hue-lb.conf.example
hue-4.11.0/tools/load-balancer/etc/supervisor.d/haproxy.conf
hue-4.11.0/tools/load-balancer/etc/supervisor.d/nginx.conf
hue-4.11.0/tools/load-balancer/etc/supervisor.d/monitor-hue-lb.conf
hue-4.11.0/tools/load-balancer/bin/monitor-hue-lb
hue-4.11.0/tools/load-balancer/bin/supervisord
hue-4.11.0/tools/load-balancer/bin/haproxy-wrapper
hue-4.11.0/tools/load-balancer/bin/supervisorctl
hue-4.11.0/tools/app_reg/build.py
hue-4.11.0/tools/app_reg/ptb.py
hue-4.11.0/tools/app_reg/registry.py
hue-4.11.0/tools/app_reg/__pycache__/
hue-4.11.0/tools/app_reg/common.py
hue-4.11.0/tools/app_reg/app_reg.py
hue-4.11.0/tools/virtual-bootstrap/virtualenv_support/
hue-4.11.0/tools/virtual-bootstrap/virtual-bootstrap.py
hue-4.11.0/tools/virtual-bootstrap/virtualenv_support/__init__.py
hue-4.11.0/tools/virtual-bootstrap/virtualenv_support/pip-9.0.1-py2.py3-none-any.whl
hue-4.11.0/tools/virtual-bootstrap/virtualenv_support/argparse-1.4.0-py2.py3-none-any.whl
hue-4.11.0/tools/virtual-bootstrap/virtualenv_support/wheel-0.29.0-py2.py3-none-any.whl
hue-4.11.0/tools/virtual-bootstrap/virtualenv_support/setuptools-28.8.0-py2.py3-none-any.whl
root@nodo1:/#
```

- Entramos a la carpeta: cd hue-4.11.0

```
root@nodo1:/# cd hue-4.11.0
root@nodo1:/hue-4.11.0#
```

- Iniciamos la compilación: PREFIX=/opt/hadoop PYTHON\_VER=python3.8  
SYS\_PYTHON=python3.8 make install

```
Traceback (most recent call last):
 File "./build/env/bin/hue", line 33, in <module>
 sys.exit(load_entry_point('desktop', 'console_scripts', 'hue')())
 File "./build/env/bin/hue", line 25, in importlib_load_entry_point
 return next(matches).load()
 File "/usr/lib/python3.8/importlib/metadata.py", line 77, in load
 module = import_module(match.group('module'))
 File "/usr/lib/python3.8/importlib/_init_.py", line 127, in import_module
 return _bootstrap._gcd_import(name[level:], package, level)
 File "<frozen importlib._bootstrap>", line 1014, in _gcd_import
 File "<frozen importlib._bootstrap>", line 991, in _find_and_load
 File "<frozen importlib._bootstrap>", line 961, in _find_and_load_unlocked
 File "<frozen importlib._bootstrap>", line 219, in _call_with_frames_removed
 File "<frozen importlib._bootstrap>", line 1014, in _gcd_import
 File "<frozen importlib._bootstrap>", line 991, in _find_and_load
 File "<frozen importlib._bootstrap>", line 975, in _find_and_load_unlocked
 File "<frozen importlib._bootstrap>", line 671, in _load_unlocked
 File "<frozen importlib._bootstrap_external>", line 848, in exec_module
 File "<frozen importlib._bootstrap>", line 219, in _call_with_frames_removed
 File "/opt/hadoop/hue/desktop/core/src/desktop/_init_.py", line 19, in <module>
 import desktop.signal_handlers
 File "/opt/hadoop/hue/desktop/core/src/desktop/signal_handlers.py", line 20, in <module>
 from desktop.lib.thread_util import dump_traceback
 File "/opt/hadoop/hue/desktop/core/src/desktop/lib/thread_util.py", line 19, in <module>
 from future import standard_library
ModuleNotFoundError: No module named 'future'
make[1]: *** [Makefile:193: create-static] Error 1
make[1]: Leaving directory '/opt/hadoop/hue'
make: *** [Makefile:175: install-env] Error 2
root@nod01:~# /opt/hadoop/hue/build/env/bin/pip install future daemon python-daemon pytz
Collecting future
 Using cached future-1.0.0-py3-none-any.whl (491 kB)
Collecting daemon
 Using cached daemon-1.2.tar.gz (2.1 kB)
 Preparing metadata (setup.py) ... done
Collecting python-daemon
 Using cached python_daemon-3.1.2-py3-none-any.whl (30 kB)
Collecting pytz
 Using cached pytz-2026.1.post1-py2.py3-none-any.whl (510 kB)
Collecting lockfile >=0.10
 Using cached lockfile-0.12.2-py2.py3-none-any.whl (13 kB)
Building wheels for collected packages: daemon
 Building wheel for daemon (setup.py) ... done
 Created wheel for daemon: filename=daemon-1.2-py3-none-any.whl size=2738 sha256=af4c378f3ce6f038e11a55ef483411e1e9922055349908894bfdd0d7c8cc20a9
 Stored in directory: /root/.cache/pip/wheels/74/9a/af/3787ccf627273985cf706f143316d0aa49f358995bbce1405
Successfully built daemon
Installing collected packages: pytz, lockfile, daemon, python-daemon, future
```

#### **Paso 4: Corrección de Errores (Fixes)**

Antes de finalizar la instalación hemos descargado diferentes librerías que no tenemos instaladas.

Utilizando el comando: `/opt/hadoop/hue/build/env/bin/pip install`

`*nombre_de_la_libreria*` para descargar.

Y luego ejecutamos el comando `PYTHON_VER=python3.8 SYS_PYTHON=python3.8`

`make create-static` para ejecutar el contenedor.

Una vez que ejecutamos el último comando, este devuelve un `ModuleNotFoundError` si falta alguna librería o módulo y que tienes que instalar para que el programa funcione.

## Paso 5: Configuración del Ecosistema

### A. Configuración de HUE: Buscamos los siguientes valores y cambiamos sus valores

```
[[[default]]]
Enter the filesystem uri
fs_defaultfs=hdfs://nodo1:9000
```

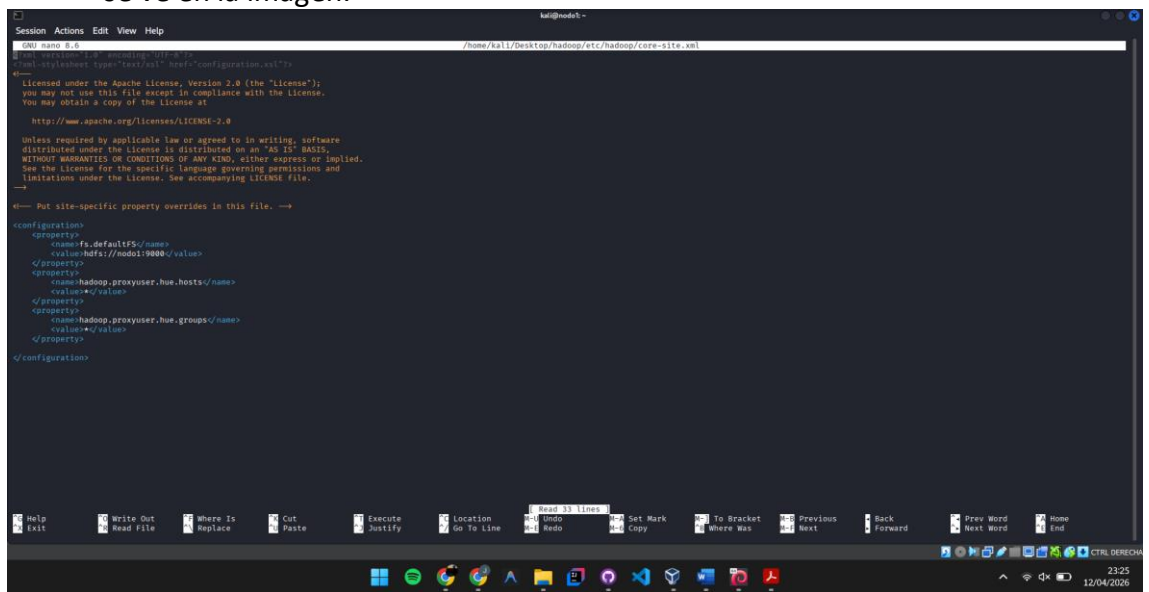
```
Use WebHdfs/HttpFs as the communication mechanism.
Domain should be the NameNode or HttpFs host.
Default port is 14000 for HttpFs.
webhdfs_url=http://nodo1:50070/webhdfs/v1
```

```
[[[default]]]
Enter the host on which you are running the ResourceManager
resourcemanager_host=nodo1
```

```
Host where HiveServer2 is running.
If Kerberos security is enabled, use fully-qualified domain name (FQDN).
hive_server_host=nodo1
```

### B. Configuración de Hadoop:

- a. En un terminal nuevo utilizando el comando: nano /home/kali/Desktop/hadoop/etc/hadoop/core-site.xml abrimos el archivo core-site.xml de hadoop y lo editamos para que muestre lo que se ve en la imagen.



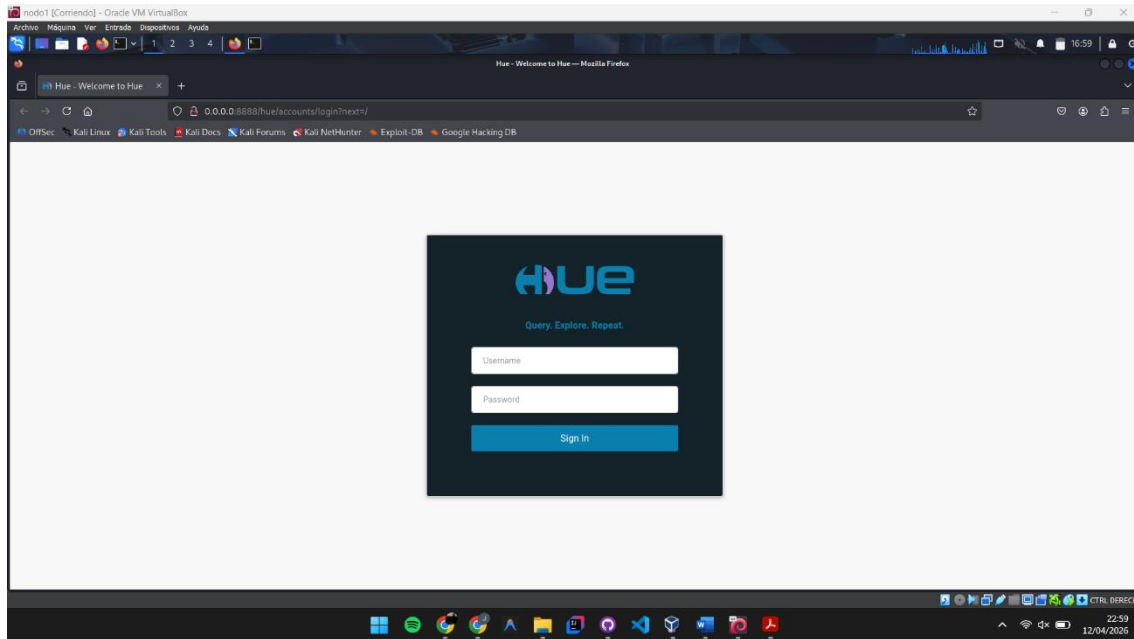
```
Session Actions Edit View Help
nan@kali:~$ nano /home/kali/Desktop/hadoop/etc/hadoop/core-site.xml
nan@kali:~$ nano /home/kali/Desktop/hadoop/etc/hadoop/core-site.xml
Licensed under the Apache License, Version 2.0 (the "license");
you may not use this file except in compliance with the License.
You may obtain a copy of the License at
http://www.apache.org/licenses/LICENSE-2.0
#
Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
#
Put site-specific property overrides in this file.
<configuration>
 <property>
 <name>fs.defaultFS</name>
 <value>hdfs://nodo1:9000</value>
 </property>
 <property>
 <name>hadoop.proxyuser.hue.hosts</name>
 <value>*.*.*.*</value>
 </property>
 <property>
 <name>hadoop.proxyuser.hue.groups</name>
 <value>*.*.*.*</value>
 </property>
</configuration>
```

## Paso 6: Arranque y Validación

Arrancamos el servidor: `su - hue -c "/opt/hadoop/hue/build/env/bin/hue runserver 0.0.0.0:8888"`

```
root@36b534a85c3e:/opt/hadoop/hue# su - hue -c "/opt/hadoop/hue/build/env/bin/hue runserver 0.0.0.0:8888"
Performing system checks ...

System check identified no issues (0 silenced).
April 12, 2026 - 13:57:58
Django version 4.2.30, using settings 'desktop.settings'
Starting development server at http://0.0.0.0:8888/
Quit the server with CONTROL-C.
```



Creamos un super usuario: `su - hue -c "/opt/hadoop/hue/build/env/bin/hue createsuperuser"`

- Nombre: Kali
- Email: (vacío)
- Password: Kali

Volvemos a la pantalla de inicio y nos registramos con ese usuario y contraseña.

